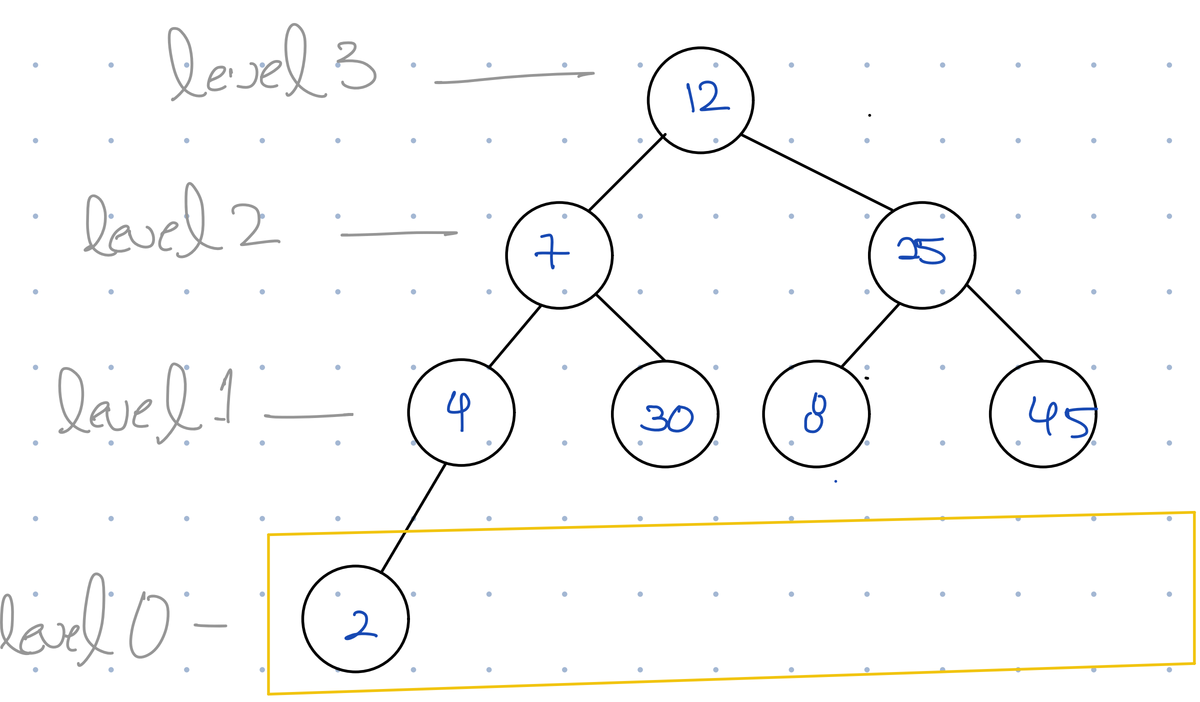
Bottom-up approach / Heapify Method

[12, 7, 25, 4, 30, 8, 45, 2]

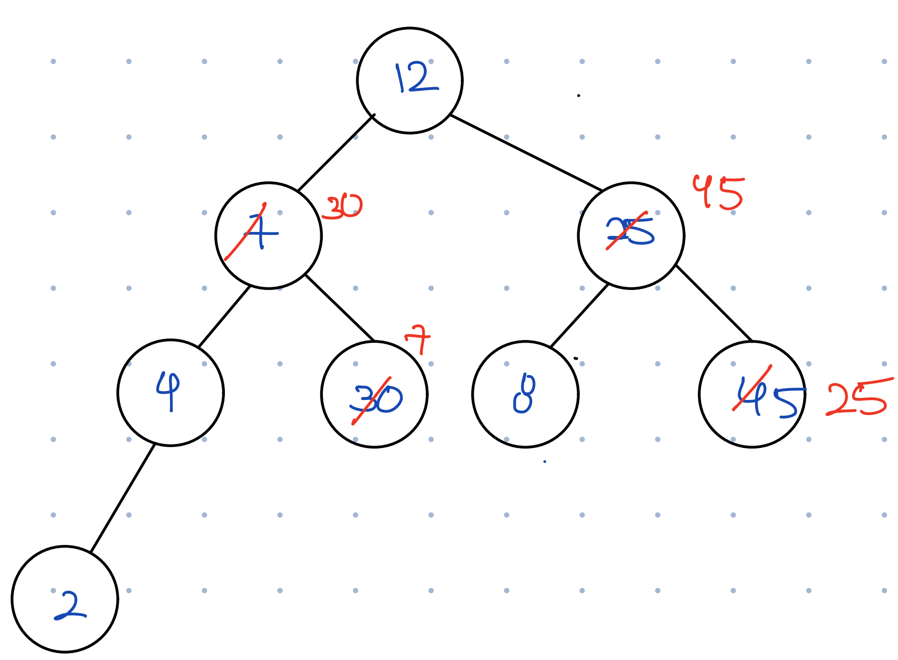
- Convert directly to a tree.

[_, 12, 7, 25, 4, 30, 8, 45, 2]
1 2 3 4 5 6 7 8

- Now, fix this to form a heap.
  - fix sub-tree at level 1, move onto level 2 & so on till level log n

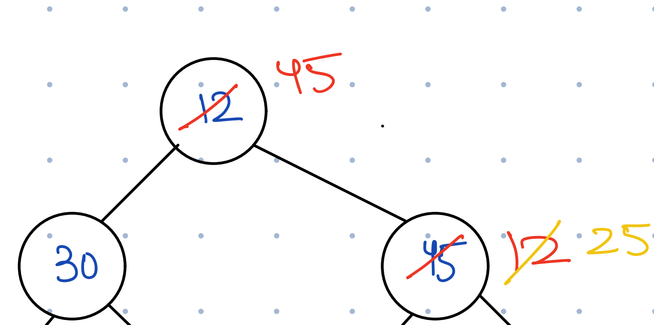


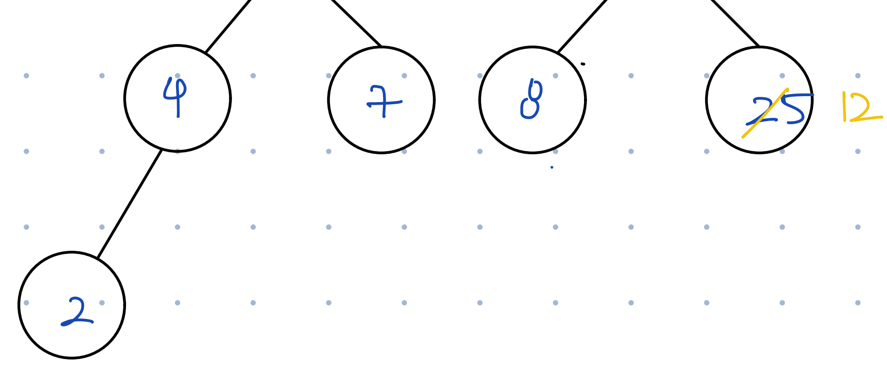
- Check all sub-trees at level 1



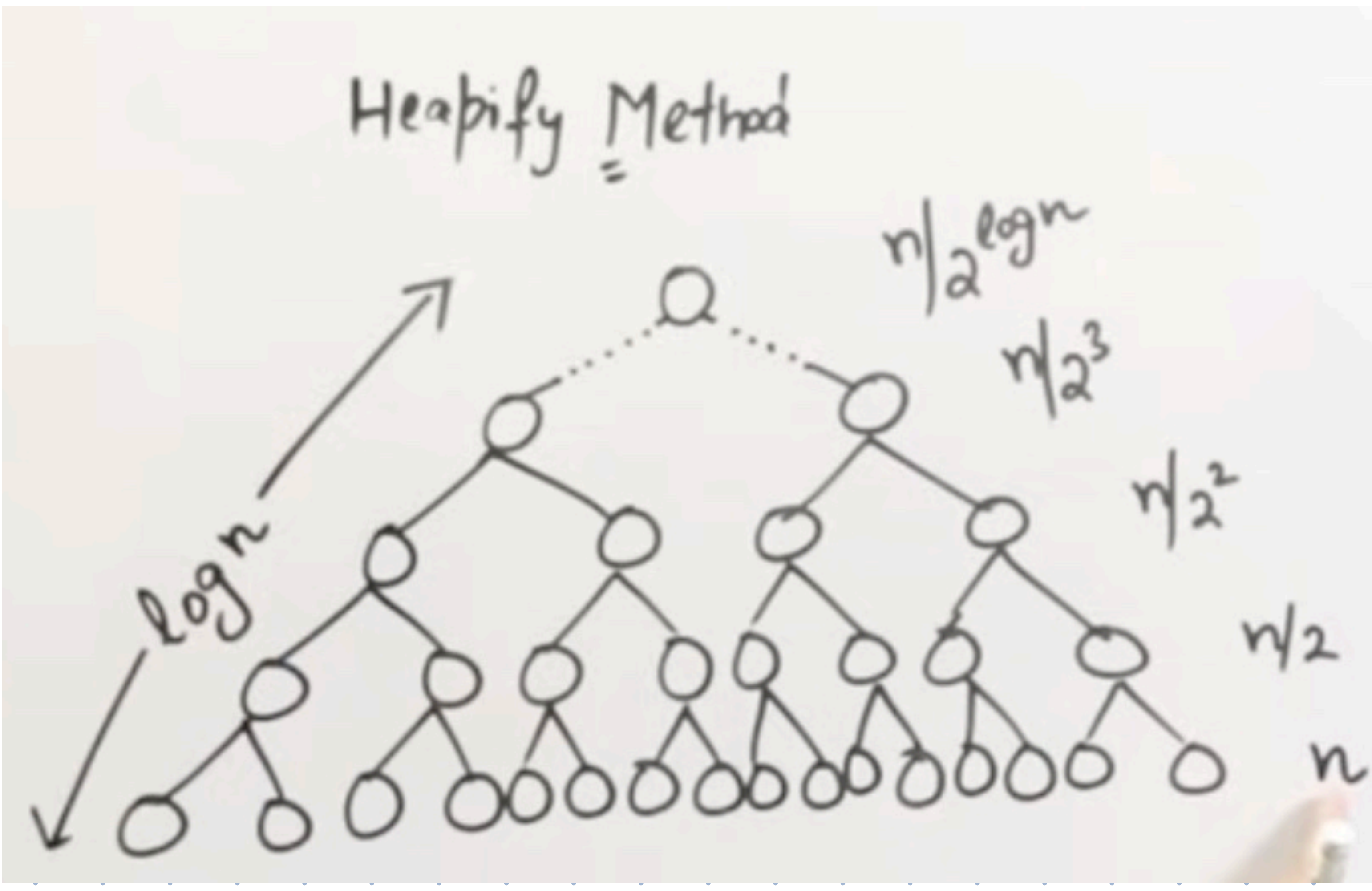
- Check all sub-trees at level 2

[_, 12, 30, 45, 4, 7, 8, 25, 2]





$$\left[\,—,\ \underset{1}{45},\ \underset{2}{30},\ \underset{3}{25},\ \underset{4}{4},\ \underset{5}{7},\ \underset{6}{8},\ \underset{7}{12},\ \underset{8}{2}\,\right]$$



Total Swaps = S

$$S = \frac{n}{2^0}*0 + \frac{n}{2^1}*1 + \frac{n}{2^2}*2 + \frac{n}{2^3}*3 \cdots + \frac{n}{2^{\log n}}*\log n$$

$$S = n\left[\frac{1}{2}+\frac{2}{2^2}+\frac{3}{2^3}+\frac{4}{2^4}+\cdots+\frac{\log n}{2^{\log n}}\right] \cdots (1)$$

Multiply eqn 1 by 1/2

$$\frac{S}{2} = n\left[\frac{1}{2^2}+\frac{2}{2^3}+\frac{3}{2^4}+\frac{4}{2^5}+\cdots+\frac{\log n-1}{2^{\log n}}+\frac{\log n}{2^{\log n+1}}\right]\cdots(2)$$

Subtract (2) from (1)

$$\frac{\log n}{2^{\log n}} - \left[\frac{\log n-1}{2^{\log n}}\right]$$

$$\frac{S}{2} = n\left[\left[\frac{1}{2}+\frac{1}{2^2}+\frac{1}{2^3}+\frac{1}{2^4}+\cdots\frac{1}{2^{\log n}}\right]-\frac{\log n}{2^{\log n+1}}\right]$$

Use GP formula $r<1$

$$\boxed{S = \frac{a\,(1-r^n)}{1-r}}$$

$$\frac{S}{2} = n\left[\left(\cancel{\frac{1}{2}}\cdot\left(1-\frac{1}{2^{\log n}}\right)\right)-\frac{\log n}{2^{\log n+1}}\right]$$

$$= n\left[\left(1 - \frac{1}{\cancel{2}^{\cancel{\log}_2 n}}\right) - \frac{\log_2 n}{2^{\log_2 n + 1}}\right]$$

$$= n\left[\left(\frac{n-1}{n}\right) - \frac{\log n}{\cancel{2}^{\cancel{\log} n} \cdot 2}\right]$$

$$= \cancel{n}\left[\frac{(n-1)}{\cancel{n}} - \frac{\log n}{\cancel{n}}\right]$$

highest order, Ignore rest

$$= n - 1 - \log n$$

$$\frac{S}{2} = n$$

$$S = 2n$$

$$O(n)$$

NOTE: heapify-up & heapify-down may create different heaps from same data.